

Question T1C1:

a) Work out exercise (2.2) from the Pang's book:

Write a program that implements the Lagrange interpolation scheme directly. Test it by evaluating $f(0.3)$ and $f(0.5)$ from the data taken from the error function with $f(0.0) = 0$, $f(0.4) = 0.428392$, $f(0.8) = 0.742101$, $f(1.2) = 0.910314$, and $f(1.6) = 0.970348$. Examine the accuracy of the interpolation by comparing the results obtained from the interpolation with the exact values $f(0.3) = 0.328627$ and $f(0.5) = 0.520500$.

b) Why and "how much" the Aitken method is faster than a direct application of the Lagrange interpolation? Estimate the computational cost (count of floating point operations) for the two methods.

c) Explain why and find an example in which direct application of Aitken method loses accuracy.